Hypothesis Testing: Calculation of Type-I Error (α) & Type-II Error (β)

May 12, 2025

Problem Statement-I

Question

A manufacturer has developed a new fishing line, which the company claims has a mean breaking strength of 15 kilograms with a standard deviation of 0.5 kilogram. To test the hypothesis that $\mu = 15$ kilograms against the alternative that $\mu < 15$ kilograms, a random sample of 50 lines will be tested. The critical region is defined to be $\bar{X} < 14.9$.

1. Find the probability of committing a type I error when H_0 is true.
2. Evaluate β for the alternatives $\mu = 14.8$ and $\mu = 14.9$ kilograms.

Given Data: We test:

$$H_0 : \mu = 15 \quad \text{vs} \quad H_1 : \mu < 15$$

$\mu = 15, \sigma = 0.5, n = 50$
Critical region: $\bar{X} < 14.9$

(a) Type-I Error:

Standard Error:

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{0.5}{\sqrt{50}} \approx 0.0707$$

Compute z-score:

$$z = \frac{14.9 - 15}{0.0707} \approx -1.414$$

Type I Error (α) $= P(\bar{x} < 14.9 \mid \mu = 15)$:

$$\alpha = P(z < -1.414) \approx 0.0785$$

Answer: $\boxed{\alpha \approx 0.0785}$

(b) Type II Error: $\mu = 14.9$

Under alternative: $\mu = 14.9$

$$z = \frac{14.9 - 14.9}{0.0707} = 0$$

$$\beta = P(\bar{X} \geq 14.9 \mid \mu = 14.9) = P(z \geq 0) = 0.5$$

Answer: $\boxed{\beta = 0.5}$

Under alternative: $\mu = 14.8$

$$z = \frac{14.9 - 14.8}{0.0707} \approx 1.414$$

$$\beta = P(\bar{X} \geq 14.9 \mid \mu = 14.8) = P(z \geq 1.414) \approx 0.0785$$

Answer: $\boxed{\beta \approx 0.0785}$

Problem Statement-II

Question

A new curing process developed for a certain type of cement results in a mean compressive strength of 5000 kilograms per square centimeter with a standard deviation of 120 kilograms. To test the hypothesis that $\mu = 5000$ against the alternative that $\mu < 5000$, a random sample of 50 pieces of cement is tested. The critical region is defined to be $\bar{X} < 4970$.

1. Find the probability of committing a type I error when H0 is true.
2. Evaluate β for the alternatives $\mu = 4970$ and $\mu = 4960$.

Given Data: We test:

$$H_0 : \mu = 5000 \quad \text{vs} \quad H_1 : \mu < 5000$$

$\mu = 5000, \sigma = 120, n = 50$
Critical region: $\bar{x} < 4960$

(a) Type-I Error:

Standard Error:

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{120}{\sqrt{50}} \approx 16.97$$

z-score:

$$z = \frac{4970 - 5000}{16.97} \approx -1.767$$

Type I error (α) = $\mathrm{P}(\bar{x} < 4970 \mid \mu = 5000)$:

$$\alpha = P(z < -1.767) \approx 0.0387$$

Answer: $\boxed{\alpha \approx 0.0387}$

(b) Type II Error: $\mu = 14.9$

New mean: $\mu = 4970$

$$z = \frac{4970 - 4970}{16.97} = 0$$

$$\beta = P(\bar{x} \geq 4970 \mid \mu = 4970) = P(z \geq 0) = 0.5$$

Answer: $\boxed{\beta = 0.5}$
New mean: $\mu = 4960$

$$z = \frac{4970 - 4960}{16.97} \approx 0.589$$

$$\beta = P(z \geq 0.589) \approx 1 - 0.7224 = 0.2776$$

Answer: $\boxed{\beta \approx 0.2776}$

Hypothesis Testing in Linear Regression (Large Sample, Z-Test)

May 12, 2025

Simple Linear Regression

Model:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2)$$

Goal: Estimate parameters β_0 and β_1, and measure goodness of fit.

Motivation for Hypothesis Testing of Regression Parameters

- In regression analysis, we estimate parameters (e.g., β_0 and β_1) that describe the relationship between independent and dependent variables.
- Testing these parameters helps us assess the validity of the model and the significance of the relationships.
- **Null Hypothesis (H_0)**: The parameter is equal to a specific value, usually zero, meaning no effect or no relationship.
- **Alternative Hypothesis (H_1)**: The parameter is not equal to the null value, indicating the existence of a significant relationship.
- By rejecting or failing to reject the null hypothesis, we can determine whether the parameter is statistically significant.
- This helps in model interpretation, making decisions, and understanding which variables play a role in explaining the variation in the response variable.

Hypothesis Testing for Regression Parameters (Z-test)

- For large samples ($n \geq 30$), we use the Z-test to test hypotheses about the regression parameters.
- The Z-statistic for testing a parameter β_j is calculated as:

$$Z = \frac{\hat{\beta}_j - \beta_j}{SE(\hat{\beta}_j)}$$

where:

- $\hat{\beta}_j$ is the estimated regression coefficient.
- β_j is the hypothesized value of the parameter under the null hypothesis (usually 0).
- $SE(\hat{\beta}_j)$ is the standard error of the estimated regression coefficient.

Hypothesis Testing for Regression Parameters (Z-test)

- The standard error $SE(\hat{\beta}_j)$ is given by:

$$SE(\hat{\beta}_j) = \frac{\sqrt{\sum(y_i - \hat{y}_i)^2/(n-k)}}{\sqrt{\sum(x_i - \bar{x})^2}} = \sqrt{\hat{\sigma}^2 \cdot \frac{1}{\sum(x_i - \bar{x})^2}}$$

  where:

  - y_i are the observed values.
  - $\hat{y}_i$ are the predicted values from the regression.
  - n is the number of data points, and k is the number of predictors (including the intercept).
  - $\hat{\sigma}^2 = \frac{(y_i - \hat{y}_i)^2}{n-k}$ is the variance of the error term.
- We compare the computed Z-value to the critical value from the standard normal distribution to determine whether to reject the null hypothesis.

Numerical Example: Data

Given:

Observation	X	Y
1	1	2
2	2	3
3	3	5
4	4	4
5	5	6

Calculating Estimates

Step 1: Calculate means

$$\bar{X} = 3, \quad \bar{Y} = 4$$

Step 2: Estimate β_1

$$\hat{\beta}_1 = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{\sum(X_i - \bar{X})^2} = \frac{9}{10} = 0.9$$

Step 3: Estimate β_0

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1\bar{X} = 4 - 0.9(3) = 1.3$$

Calculating RSS and SE

Step 4: Residuals and RSS

- Predicted values: $\hat{Y}_i = 1.3 + 0.9X_i$
- Residuals: $e_i = Y_i - \hat{Y}_i$

$$\hat{Y} = [2.2, 3.1, 4.0, 4.9, 5.8], \quad e = [-0.2, -0.1, 1.0, -0.9, 0.2]$$

$$RSS = \sum e_i^2 = 0.04 + 0.01 + 1.00 + 0.81 + 0.04 = 1.90$$

Step 5: Estimate σ^2

$$\hat{\sigma}^2 = \frac{RSS}{n-2} = \frac{1.90}{3} \approx 0.633$$

Standard Error of $\hat{\beta}_1$

$$SE(\hat{\beta}_1) = \sqrt{\hat{\sigma}^2 \cdot \frac{1}{\sum(X_i - \bar{X})^2}} = \sqrt{0.633 \cdot \frac{1}{10}} \approx 0.252$$

Conclusion: Standard error of the slope estimate is approximately 0.252.

Simple Linear Regression Example (Large Sample)

Scenario: Predict annual income based on years of education for 1500 adults.
Model: $Y = \beta_0 + \beta_1 X + \varepsilon$
Regression Output:

Coefficient	Estimate	Standard Error
β_0 (Intercept)	10,000	—
β_1 (Years of Education)	2,000	150

Hypothesis: $H_0 : \beta_1 = 0$ vs $H_1 : \beta_1 > 0$ (right-tailed)

$$z = \frac{2000 - 0}{150} = 13.33$$

Decision: $13.33 > 1.645 \Rightarrow$ Reject H_0 at $\alpha = 0.05$

Hypothesis Testing in Multiple Regression (Large Sample, Z-Test)

May 12, 2025

General Regression Model

Model:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2)$$

Z-test for Coefficient:

$$H_0 : \beta_j = \beta_{j0} \quad \text{vs} \quad H_1 : \beta_j \neq \beta_{j0}$$

$$z = \frac{\hat{\beta}_j - \beta_{j0}}{SE(\hat{\beta}_j)}$$

Example 1: Predicting House Price

Scenario: Model house price based on size and bedrooms using 1000 houses.

Regression Output:

Coefficient	Estimate	Standard Error
β_0 (Intercept)	50,000	—
β_1 (Size)	120	10
β_2 (Bedrooms)	5,000	1,000

Test: $H_0 : \beta_1 = 0$ vs $H_1 : \beta_1 \neq 0$

$$z = \frac{120 - 0}{10} = 12.0$$

Decision: $|z| > 1.96 \Rightarrow$ Reject H_0.

Example 2: Predicting Employee Salary

Scenario: Model salary based on experience and certifications for 500 employees.

Regression Output:

Coefficient	Estimate	Standard Error
β_0	30,000	—
β_1 (Experience)	1,500	200
β_2 (Certifications)	3,000	1,200

Test: $H_0 : \beta_2 = 0$ vs $H_1 : \beta_2 \neq 0$

$$z = \frac{3000 - 0}{1200} = 2.5$$

Decision: $2.5 < 2.576 \Rightarrow$ Fail to reject H_0 at $\alpha = 0.01$.

Example 3: Predicting Student GPA

Scenario: Model GPA using SAT score and high school GPA for 800 students.

Regression Output:

Coefficient	Estimate	Standard Error
β_0	0.5	—
β_1 (SAT)	0.002	0.0004
β_2 (HS GPA)	0.4	0.1

Test: $H_0 : \beta_2 = 0.5$ vs $H_1 : \beta_2 < 0.5$ (left-tailed)

$$z = \frac{0.4 - 0.5}{0.1} = -1.0$$

Decision: $-1.0 > -1.645 \Rightarrow$ Fail to reject H_0 at $\alpha = 0.05$.

Numerical Example: Calculating SE of $\hat{\beta}_1$

Given: $n = 4, p = 3$ (including intercept)
Data:

Obs	Y	X_1	X_2
1	5	1	4
2	6	2	5
3	7	3	6
4	8	4	7

Step 1: Matrices

$$X = \begin{bmatrix} 1 & 1 & 4 \\ 1 & 2 & 5 \\ 1 & 3 & 6 \\ 1 & 4 & 7 \end{bmatrix}, \quad Y = \begin{bmatrix} 5 \\ 6 \\ 7 \\ 8 \end{bmatrix}$$

Numerical Example (contd.)

Step 2: Compute $\hat{\beta}$

$$\hat{\beta} = (X^\top X)^{-1} X^\top Y$$

Assume:

$$\hat{\beta} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

Step 3: Compute Residuals and RSS

$$\hat{Y} = X\hat{\beta} = \begin{bmatrix} 3 \\ 4 \\ 5 \\ 6 \end{bmatrix}, \quad e = Y - \hat{Y} = \begin{bmatrix} 2 \\ 2 \\ 2 \\ 2 \end{bmatrix}$$

$$RSS = e^\top e = 2^2 + 2^2 + 2^2 + 2^2 = 16$$

$$\hat{\sigma}^2 = \frac{16}{4-3} = 16$$

Numerical Example (Final Step)

Step 4: Compute SE of $\hat{\beta}_1$

$$SE(\hat{\beta}_1) = \sqrt{\hat{\sigma}^2 \cdot [(X^\top X)^{-1}]_{22}} = \sqrt{16 \cdot 0.5} = \sqrt{8} \approx 2.828$$

Conclusion: Estimated standard error of $\hat{\beta}_1$ is approximately 2.828.

Standard Error in Multiple Regression

May 12, 2025

Multiple Linear Regression Model

Model:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_{p-1} X_{p-1,i} + \varepsilon_i$$

where:

- Y_i is the response variable
- X_{ji} are predictor variables
- $\varepsilon_i \sim N(0, \sigma^2)$ are errors

Matrix form:

$$Y = X\beta + \varepsilon$$

Interpretation and Estimation of $\hat{\beta}$

What is $\hat{\beta}$?

- It is the vector of estimated regression coefficients.
- It minimizes the sum of squared residuals.

Formula:

$$\hat{\beta} = (X^\top X)^{-1} X^\top Y$$

Interpretation: Each $\hat{\beta}_j$ represents the estimated change in Y for a one-unit increase in X_j, holding other variables constant.

Numerical Example: Dataset

Given: $n = 4$ observations, $p = 3$ parameters (including intercept)
Data:

Obs	Y	X_1	X_2
1	5	1	4
2	6	2	5
3	7	3	6
4	8	4	7

Construct the design matrix:

$$X = \begin{bmatrix} 1 & 1 & 4 \\ 1 & 2 & 5 \\ 1 & 3 & 6 \\ 1 & 4 & 7 \end{bmatrix}, \quad Y = \begin{bmatrix} 5 \\ 6 \\ 7 \\ 8 \end{bmatrix}$$

Step 1: Compute $\hat{\beta}$

Use: $\hat{\beta} = (X^\top X)^{-1} X^\top Y$
Assume (after matrix multiplication):

$$X^\top X = \begin{bmatrix} 4 & 10 & 22 \\ 10 & 30 & 70 \\ 22 & 70 & 174 \end{bmatrix}, \quad X^\top Y = \begin{bmatrix} 26 \\ 70 \\ 166 \end{bmatrix}$$

$$(X^\top X)^{-1} = \begin{bmatrix} 7 & -4 & 1 \\ -4 & 3 & -1 \\ 1 & -1 & 0.5 \end{bmatrix}$$

$$\hat{\beta} = (X^\top X)^{-1} X^\top Y = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

Step 2: Residuals and $\hat{\sigma}^2$

Predicted Y:

$$\hat{Y} = X\hat{\beta} = \begin{bmatrix} 3 \\ 4 \\ 5 \\ 6 \end{bmatrix} \quad \Rightarrow \quad e = Y - \hat{Y} = \begin{bmatrix} 2 \\ 2 \\ 2 \\ 2 \end{bmatrix}$$

Residual Sum of Squares (RSS):

$$RSS = e^\top e = 2^2 + 2^2 + 2^2 + 2^2 = 16$$

Estimate of variance:

$$\hat{\sigma}^2 = \frac{RSS}{n-p} = \frac{16}{4-3} = 16$$

Step 3: Standard Error of $\hat{\beta}_1$

Use:

$$SE(\hat{\beta}_j) = \sqrt{\hat{\sigma}^2 \cdot [(X^\top X)^{-1}]_{jj}}$$

For $j = 1$:

$$SE(\hat{\beta}_1) = \sqrt{16 \cdot 3} = \sqrt{48} \approx 6.93$$

Interpretation: The standard error reflects the variability in $\hat{\beta}_1$ due to sampling error.

Logistic Regression

May 12, 2025

Introduction to the Logistic Function

Logistic Function:

$$f(z) = \frac{1}{1 + e^{-z}}$$

Key Properties:

- S-shaped (sigmoid) curve
- Output always between 0 and 1
- $f(z) \to 1$ as $z \to +\infty$, and $f(z) \to 0$ as $z \to -\infty$
- Symmetric around $z = 0$: $f(0) = 0.5$

In Logistic Regression:

$$P(Y = 1 \mid X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p)}}$$

This ensures predicted probabilities remain between 0 and 1.

Introduction to Logistic Regression

Purpose: Logistic regression is used to model the probability of a binary outcome (e.g., success/failure, yes/no, 0/1) based on one or more predictor variables.

Model Form:

$$P(Y = 1 \mid X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p)}}$$

Log-Odds (Logit) Form:

$$\log\left(\frac{P(Y = 1)}{1 - P(Y = 1)}\right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

Why Use It?

- Predicts probability while keeping values in [0, 1]
- Interpretable via odds and log-odds
- Suitable for classification problems